

UNIT – 2 Python Program Flow Control

Conditional blocks:

◇ 1. Conditional Blocks (Decision Making)

Conditional statements help us to **control the flow of a program**.

They allow the program to choose between different actions depending on conditions (True/False).

☒ (a) if Statement

- Runs **only if** the condition is True.
- If condition is False → nothing happens.

```
temperature = 35
if temperature > 30:
    print("It is a hot day")
```

Output:

It is a hot day

Real Life Example:

- If traffic light is red → Stop.
- Otherwise → nothing to do.

☒ (b) if-else Statement

- If condition is True → if block runs.
- Otherwise → else block runs.

```
age = 16
if age >= 18:
    print("You can vote")
else:
    print("You cannot vote")
```

Output:

You cannot vote

Real Life Example:

- If rain → carry umbrella, else → no umbrella needed.

(c) if-elif-else Statement

- Used when we have **multiple conditions**.
- First True condition will run, others will be skipped.

```
marks = 75
if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
elif marks >= 40:
    print("Grade C")
else:
    print("Fail")
```

Output:

Grade B

Real Life Example:

- If temperature > 35 → Hot
- Else if temperature > 20 → Pleasant
- Else if temperature > 10 → Cold

- Else → Freezing

☒ (d) Nested if

- One if statement inside another.

```
age = 20
citizen = True

if age >= 18:
    if citizen:
        print("Eligible to vote")
    else:
        print("Not eligible - not a citizen")
else:
    print("Too young to vote")
```

Output:

Eligible to vote

◇ 2. For Loops

A **for loop** is used when we want to **repeat code a fixed number of times**.

☒ Using range()

- Syntax: range(start, stop, step)
- Default: start=0, step=1

```
for i in range(5):
    print(i)
```

Output:

```
0 1 2 3 4
for i in range(1, 6):
    print(i)
```

Output:

```
1 2 3 4 5
for i in range(2, 11, 2):
    print(i)
```

Output:

```
2 4 6 8 10
```

For Loop with String

```
name = "PYTHON"
for ch in name:
    print(ch)
```

Output:

```
P Y T H O N
```

For Loop with List

```
fruits = ["apple", "banana", "mango"]
for fruit in fruits:
    print(fruit)
```

Output:

```
apple
banana
```

mango

☑ For Loop with Dictionary

```
student = {"name": "Gautam", "age": 20, "course": "BTech"}
for key, value in student.items():
    print(key, ":", value)
```

Output:

```
name : Gautam
age : 20
course : BTech
```

☑ For Loop with Nested Structures

```
matrix = [[1, 2], [3, 4], [5, 6]]
for row in matrix:
    for num in row:
        print(num, end=" ")
```

Output:

```
1 2 3 4 5 6
```

◇ 3. While Loops

- Runs until condition becomes False.
- Used when **we don't know how many times loop will run**.

```
count = 1
while count <= 5:
    print("Count is:", count)
```

```
count += 1
```

Output:

```
Count is: 1
Count is: 2
Count is: 3
Count is: 4
Count is: 5
```

Infinite Loop Example

```
while True:
    print("This will run forever")
```

 Must use break inside to stop.

While Loop with User Input

```
password = ""
while password != "1234":
    password = input("Enter password: ")
print("Access Granted")
```

◇ 4. Loop Control Statements

(a) break → stop the loop immediately

```
for i in range(1, 10):
    if i == 5:
        break
```

```
print(i)
```

Output:

1 2 3 4

☒ (b) continue → skip current iteration

```
for i in range(1, 6):  
    if i == 3:  
        continue  
    print(i)
```

Output:

1 2 4 5

☒ (c) pass → do nothing (placeholder)

- **pass** is used when we need code later but don't want errors now.

```
for i in range(5):  
    pass    # Will write code later  
print("Loop finished")  
age = 20  
if age >= 18:  
    pass    # Placeholder  
else:  
    print("Not eligible")
```

👉 Without pass, Python will give an **IndentationError**.

☑ (d) else with loop

- Runs when loop finishes normally (without break).

```
for i in range(1, 5):  
    print(i)  
else:  
    print("Loop finished")
```

Output:

```
1 2 3 4  
Loop finished
```

◇ 5. Programming Examples

Example 1: Print Even Numbers (1–10)

```
for i in range(1, 11):  
    if i % 2 == 0:  
        print(i)
```

Output:

```
2 4 6 8 10
```

Example 2: Factorial using While Loop

```
n = 5  
fact = 1  
while n > 0:  
    fact *= n  
    n -= 1
```



```
print("Factorial:", fact)
```

Output:

Factorial: 120

Example 3: Search in a List

```
fruits = ["apple", "banana", "mango"]
search = "banana"
for fruit in fruits:
    if fruit == search:
        print("Found:", fruit)
        break
else:
    print("Not found")
```

Output:

Found: banana

Example 4: Multiplication Table

```
num = 5
for i in range(1, 11):
    print(num, "x", i, "=", num*i)
```

Output:

```
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

Example 5: Check Prime Number

```
num = 7
is_prime = True

for i in range(2, num):
    if num % i == 0:
        is_prime = False
        break

if is_prime:
    print(num, "is Prime")
else:
    print(num, "is Not Prime")
```

Output:

7 is Prime

Example 6: Sum of Numbers in List

```
numbers = [10, 20, 30, 40]
total = 0
for n in numbers:
    total += n
print("Sum =", total)
```

Output:

Sum = 100

Example 7: Reverse String using Loop

```
word = "PYTHON"
reverse = ""
for ch in word:
```

```
reverse = ch + reverse
print("Reverse:", reverse)
```

Output:

Reverse: NOHTYP

◇ Difference Table Questions (with Solutions)

1. Difference between if and if-else

Point	if	if-else
Meaning	Runs a block of code only if condition is True	Runs one block if condition True, otherwise another block
Output	May or may not execute code	Always executes one block
Example	if marks >= 40: print("Pass")	if marks >= 40: print("Pass") else: print("Fail")

✓ Example Code:

```
marks = 30
```

```
# using if
if marks >= 40:
    print("Pass")
```

```
# using if-else
if marks >= 40:
    print("Pass")
else:
    print("Fail")
```

2. Difference between for loop and while loop

Point	For Loop	While Loop
Use	When we know the number of iterations	When we don't know how many times to repeat
Syntax	for i in range(n)	while condition:
Control	Automatically updates counter	We have to manually update counter
Example	Print 1 to 5 using for	Print 1 to 5 using while

✅ Example Code:

```
# For loop
for i in range(1, 6):
    print(i)
```

```
# While loop
i = 1
while i <= 5:
    print(i)
    i += 1
```

3. Difference between break and continue

Point	break	continue
Use	Exits the loop completely	Skips current iteration, continues loop
Effect	Loop ends immediately	Only skips one step, loop continues
Example	Stop printing when number = 3	Skip printing when number = 3

✅ Example Code:

```
# break
for i in range(1, 6):
    if i == 3:
        break
    print(i)    # Output: 1, 2
```

```
# continue
for i in range(1, 6):
    if i == 3:
        continue
```

```
        continue
print(i)    # Output: 1, 2, 4, 5
```

4. Difference between pass and continue

Point	pass	continue
Meaning	Placeholder → does nothing, just skips execution	Skips current iteration and goes to next loop
Use Case	When code is not ready but you don't want error	When you want to skip specific condition
Example	Empty if block	Skip odd numbers

✅ Example Code:

```
# pass
for i in range(1, 6):
    if i == 3:
        pass # does nothing
    print(i)  # Output: 1, 2, 3, 4, 5
```

```
# continue
for i in range(1, 6):
    if i % 2 != 0:
        continue # skips odd numbers
    print(i)     # Output: 2, 4
```

5. Difference between list and string iteration in for loop

Point	String	List
Meaning	Collection of characters	Collection of items (numbers, strings, etc.)
Iteration	Loops through characters	Loops through elements
Example	"hello" → h e l l o	[10, 20, 30] → 10 20 30

✅ Example Code:

```
# Iterating string
for ch in "hello":
    print(ch)

# Iterating list
for num in [10, 20, 30]:
    print(num)
```

6. Difference between range() and len()

Point	range()	len()
Meaning	Generates a sequence of numbers	Returns length of a collection
Use	To create loops	To count number of elements
Example	range(5) → 0,1,2,3,4	len("hello") → 5

✅ Example Code:

```
# range()
for i in range(5):
    print(i)

# len()
word = "hello"
print(len(word)) # Output: 5
```

7. Difference between == and is

Point	==	is
Meaning	Compares values	Compares memory location (identity)
Example	5 == 5 → True	5 is 5 → True (but works differently for objects)

✅ Example Code:

```
x = [1, 2, 3]
y = [1, 2, 3]
```

```
print(x == y)    # True (values same)
print(x is y)    # False (different memory)
```

Important Questions & Answers

Q1. What is flow control in Python? Why is it important?

Answer:

- Flow control means controlling **which part of the program will run** and **how many times it will run**.
- It is important because:
 - It allows decision making (if/else).
 - It allows repeating tasks (loops).
 - It makes programs dynamic and useful.

Example:

```
marks = 80
if marks >= 50:
    print("Pass")
else:
    print("Fail")
```

Q2. Explain if, else, elif statements with example.

Answer:

- **if:** runs only if condition is true.
- **else:** runs when condition is false.
- **elif:** used for multiple conditions.

Example:

```
marks = 72
if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
else:
    print("Grade C")
```

Output:

Grade B

Q3. What is a for loop in Python? Explain with example.

Answer:

- A **for loop** is used when we know how many times to repeat.
- It is often used with `range()`, lists, strings, or dictionaries.

Example with range:

```
for i in range(1, 6):
    print(i)
```

Output:

1 2 3 4 5

Q4. Explain for loop with string, list and dictionary with examples.

Answer:

- **With String**


```
for ch in "HELLO":  
    print(ch)
```

- **With List**

```
fruits = ["apple", "banana", "mango"]  
for f in fruits:  
    print(f)
```

- **With Dictionary**

```
student = {"name": "Gautam", "age": 20}  
for key, value in student.items():  
    print(key, ":", value)
```

Q5. What is a while loop? Give example.

Answer:

- A **while loop** repeats until the condition becomes False.
- It is used when we do not know the number of repetitions in advance.

Example:

```
count = 1  
while count <= 5:  
    print("Count =", count)  
    count += 1
```

Output:

```
Count = 1  
Count = 2  
Count = 3  
Count = 4  
Count = 5
```

Q6. What are break, continue and pass statements? Explain with example.

Answer:

- **break**: stops the loop immediately.

```
for i in range(1, 6):  
    if i == 3:  
        break  
    print(i)  
# Output: 1 2
```

- **continue**: skips the current iteration.

```
for i in range(1, 6):  
    if i == 3:  
        continue  
    print(i)  
# Output: 1 2 4 5
```

- **pass**: does nothing, used as placeholder.

```
for i in range(5):  
    pass  
print("Loop finished")
```

Q7. Explain else with loops in Python. Give example.

Answer:

- A loop can have an **else** block.
- It runs when loop finishes normally (not stopped by break).

Example:

```
for i in range(3):  
    print(i)
```

```
else:  
    print("Loop ended normally")
```

Output:

```
0  
1  
2  
Loop ended normally
```

Q8. Write a program to print multiplication table of 5 using for loop.

Answer:

```
for i in range(1, 11):  
    print("5 x", i, "=", 5*i)
```

Output:

```
5 x 1 = 5  
5 x 2 = 10  
...  
5 x 10 = 50
```

Q9. Write a program to check whether a number is prime or not.

Answer:

```
num = 7  
is_prime = True  
for i in range(2, num):  
    if num % i == 0:  
        is_prime = False
```

```
        break
if is_prime:
    print("Prime Number")
else:
    print("Not Prime")
```

Output:

Prime Number

Q10. Write a program using while loop to find factorial of a number.

Answer:

```
n = 5
fact = 1
while n > 0:
    fact *= n
    n -= 1
print("Factorial =", fact)
```

Output:

Factorial = 120

Q11. Write a program to print even numbers between 1 and 20.

Answer:

```
for i in range(1, 21):
    if i % 2 == 0:
        print(i)
```

Output:

2 4 6 8 10 12 14 16 18 20

Q12. Write a program to reverse a string using loop.

Answer:

```
word = "PYTHON"
reverse = ""
for ch in word:
    reverse = ch + reverse
print("Reverse =", reverse)
```

Output:

Reverse = NOHTYP

Most Important Python Coding Questions (with Solutions)

Q1. Print the multiplication table of a number.

```
num = 7
for i in range(1, 11):
    print(num, "x", i, "=", num*i)
```

Output:

```
7 x 1 = 7
7 x 2 = 14
...
```

$$7 \times 10 = 70$$

Q2. Find factorial of a number using while loop.

```
n = 5
fact = 1
while n > 0:
    fact *= n
    n -= 1
print("Factorial =", fact)
```

Output:

Factorial = 120

Q3. Check whether a number is prime or not.

```
num = 13
is_prime = True

for i in range(2, num):
    if num % i == 0:
        is_prime = False
        break

if is_prime:
    print(num, "is Prime")
else:
    print(num, "is Not Prime")
```

Output:

13 is Prime

Q4. Print even numbers between 1 and 20.

```
for i in range(1, 21):  
    if i % 2 == 0:  
        print(i)
```

Output:

2 4 6 8 10 12 14 16 18 20

Q5. Reverse a string using for loop.

```
word = "PYTHON"  
reverse = ""  
for ch in word:  
    reverse = ch + reverse  
print("Reverse =", reverse)
```

Output:

Reverse = NOHTYP

Q6. Count vowels in a string.

```
text = "hello world"  
vowels = "aeiou"  
count = 0
```

```
for ch in text:  
    if ch in vowels:  
        count += 1
```

```
print("Number of vowels =", count)
```

Output:

Number of vowels = 3

Q7. Find the sum of numbers in a list.

```
numbers = [10, 20, 30, 40, 50]  
total = 0
```

```
for n in numbers:  
    total += n
```

```
print("Sum =", total)
```

Output:

Sum = 150

Q8. Print Fibonacci series up to n terms.

```
n = 10  
a, b = 0, 1  
count = 0  
  
while count < n:  
    print(a, end=" ")  
    a, b = b, a+b  
    count += 1
```

Output:

0 1 1 2 3 5 8 13 21 34

Q9. Search an element in a list.

```
fruits = ["apple", "banana", "mango", "orange"]
search = "banana"
```

```
for f in fruits:
    if f == search:
        print("Found:", f)
        break
else:
    print("Not Found")
```

Output:

Found: banana

Q10. Check whether a number is palindrome or not.

(A number is palindrome if it reads same forward and backward, e.g., 121)

```
num = 121
temp = num
reverse = 0
```

```
while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num //= 10
```

```
if temp == reverse:
    print("Palindrome Number")
else:
    print("Not Palindrome")
```

Output:

Palindrome Number

Q11. Print numbers from 1 to 10, skip 5 using continue.

```
for i in range(1, 11):  
    if i == 5:  
        continue  
    print(i)
```

Output:

1 2 3 4 6 7 8 9 10

Q12. Print the largest number in a list.

```
numbers = [15, 88, 23, 45, 92, 10]  
largest = numbers[0]
```

```
for n in numbers:  
    if n > largest:  
        largest = n
```

```
print("Largest number =", largest)
```

Output:

Largest number = 92

Important Python Coding Questions with Solutions

1. Multiplication Table of a Number

```
num = 7
for i in range(1, 11):
    print(num, "x", i, "=", num*i)
```

Output:

```
7 x 1 = 7
7 x 2 = 14
...
7 x 10 = 70
```

2. Factorial of a Number (using while loop)

```
n = 5
fact = 1
while n > 0:
    fact *= n
    n -= 1
print("Factorial =", fact)
```

Output:

```
Factorial = 120
```

3. Prime Number Check

```
num = 13
is_prime = True

for i in range(2, num):
    if num % i == 0:
        is_prime = False
        break

if is_prime:
    print(num, "is Prime")
else:
    print(num, "is Not Prime")
```

Output:

13 is Prime

4. Print Even Numbers (1 to 20)

```
for i in range(1, 21):
    if i % 2 == 0:
        print(i)
```

Output:

2 4 6 8 10 12 14 16 18 20

5. Reverse a String

```
word = "PYTHON"
reverse = ""
for ch in word:
    reverse = ch + reverse
```

```
print("Reverse =", reverse)
```

Output:

```
Reverse = NOHTYP
```

6. Count Vowels in String

```
text = "hello world"  
vowels = "aeiou"  
count = 0
```

```
for ch in text:  
    if ch in vowels:  
        count += 1
```

```
print("Number of vowels =", count)
```

Output:

```
Number of vowels = 3
```

7. Sum of Numbers in a List

```
numbers = [10, 20, 30, 40, 50]  
total = 0
```

```
for n in numbers:  
    total += n
```

```
print("Sum =", total)
```

Output:

Sum = 150

8. Fibonacci Series (first n terms)

```
n = 10
a, b = 0, 1
count = 0

while count < n:
    print(a, end=" ")
    a, b = b, a+b
    count += 1
```

Output:

0 1 1 2 3 5 8 13 21 34

9. Search Element in a List

```
fruits = ["apple", "banana", "mango", "orange"]
search = "banana"

for f in fruits:
    if f == search:
        print("Found:", f)
        break
else:
    print("Not Found")
```

Output:

Found: banana

10. Palindrome Number Check

```
num = 121
temp = num
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num //= 10

if temp == reverse:
    print("Palindrome Number")
else:
    print("Not Palindrome")
```

Output:

Palindrome Number

11. Use of Continue (skip number 5)

```
for i in range(1, 11):
    if i == 5:
        continue
    print(i)
```

Output:

1 2 3 4 6 7 8 9 10

12. Find Largest Number in a List

```
numbers = [15, 88, 23, 45, 92, 10]
largest = numbers[0]
```

```
for n in numbers:
    if n > largest:
        largest = n

print("Largest number =", largest)
```

Output:

Largest number = 92

13. Find Smallest Number in a List

```
numbers = [15, 88, 23, 45, 92, 10]
smallest = numbers[0]

for n in numbers:
    if n < smallest:
        smallest = n

print("Smallest number =", smallest)
```

Output:

Smallest number = 10

14. Find Sum of Digits of a Number

```
num = 1234
total = 0

while num > 0:
    digit = num % 10
    total += digit
    num //= 10
```



```
print("Sum of digits =", total)
```

Output:

Sum of digits = 10

15. Armstrong Number Check

(A number is Armstrong if sum of cubes of digits = number itself, e.g., 153)

```
num = 153
temp = num
total = 0

while num > 0:
    digit = num % 10
    total += digit ** 3
    num //= 10

if total == temp:
    print("Armstrong Number")
else:
    print("Not Armstrong")
```

Output:

Armstrong Number

16. Print Numbers 1 to 10 using while loop

```
i = 1
while i <= 10:
    print(i)
```

```
i += 1
```

17. Print Odd Numbers Between 1 and 20

```
for i in range(1, 21):  
    if i % 2 != 0:  
        print(i)
```

18. Calculate Average of List

```
numbers = [10, 20, 30, 40, 50]  
total = 0
```

```
for n in numbers:  
    total += n
```

```
avg = total / len(numbers)  
print("Average =", avg)
```

Output:

Average = 30.0

19. Print First 10 Natural Numbers

```
for i in range(1, 11):  
    print(i)
```

20. Print Factor of a Number

```
num = 12
print("Factors of", num, ":")
for i in range(1, num+1):
    if num % i == 0:
        print(i)
```

Output:

```
Factors of 12 :
1 2 3 4 6 12
```